Security and privacy review

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask where content enters context and which actor may read it.

Teach scoped confidentiality tests without real private records or complete-audit overclaims.

Keep one habit: Model direct/cache mismatch → gate before lookup → inspect assigned-note access/counts → assess export bypass → record untested paths.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of security and **privacy review**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Performance and cost experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of performance and cost experiments

Reference example, not the only acceptable wording: Teacher explanation: Performance experiments compare matched workloads with declared timing boundaries and quality constraints. Averages can hide slow cases.

Use the same cases and environment; record every duration and success. For a simple median sort values and average the middle pair for even counts. Keep failed requests visible.

Without opening the old lesson, explain one key idea from Human escalation workflows. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of human escalation workflows

Reference example, not the only acceptable wording: Teacher explanation: Escalation routes unresolved decisions to an authorised person with sufficient context. It must define owner, deadline and fallback.

Toy workflow routes conflicting room approval to coordinator with source IDs and requested decision. While awaiting review the assistant returns needs_review and performs no booking.

Without opening the old lesson, explain one key idea from Release and rollback. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of release and rollback

Reference example, not the only acceptable wording: Teacher explanation: Release records bind code, configuration, data schema and evaluation **evidence**. Rollback requires compatibility with the current stored data.

Version v2 writes schema 2. Version v1 reads only schema 1. A rollback to v1 after v2 writes data cannot safely proceed without migration or a compatible reader.

Pictured end goal

Explain security and **privacy review** using the deciding step in this fictional service.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Performance and cost experiments

Human escalation workflows

Release and rollback

Use the three-session teaching cycle above, with extra time as needed. Security and **privacy review** checks assets, actors, data flows and permissions against explicit threats. A clean happy path does not test misuse. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of security and **privacy review**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach confidentiality across paths and minimized secondary use

> Inspect content entry into context as well as final output.

Definitions and complete fictional access contract

Security and **privacy review** checks assets, actors, data flows and permissions against explicit threats. A clean happy path does not test misuse.

For the classroom service identify restricted guides, session secrets and student content. Test denied roles, unexpected fields, source injection and minimised logging without real credentials.

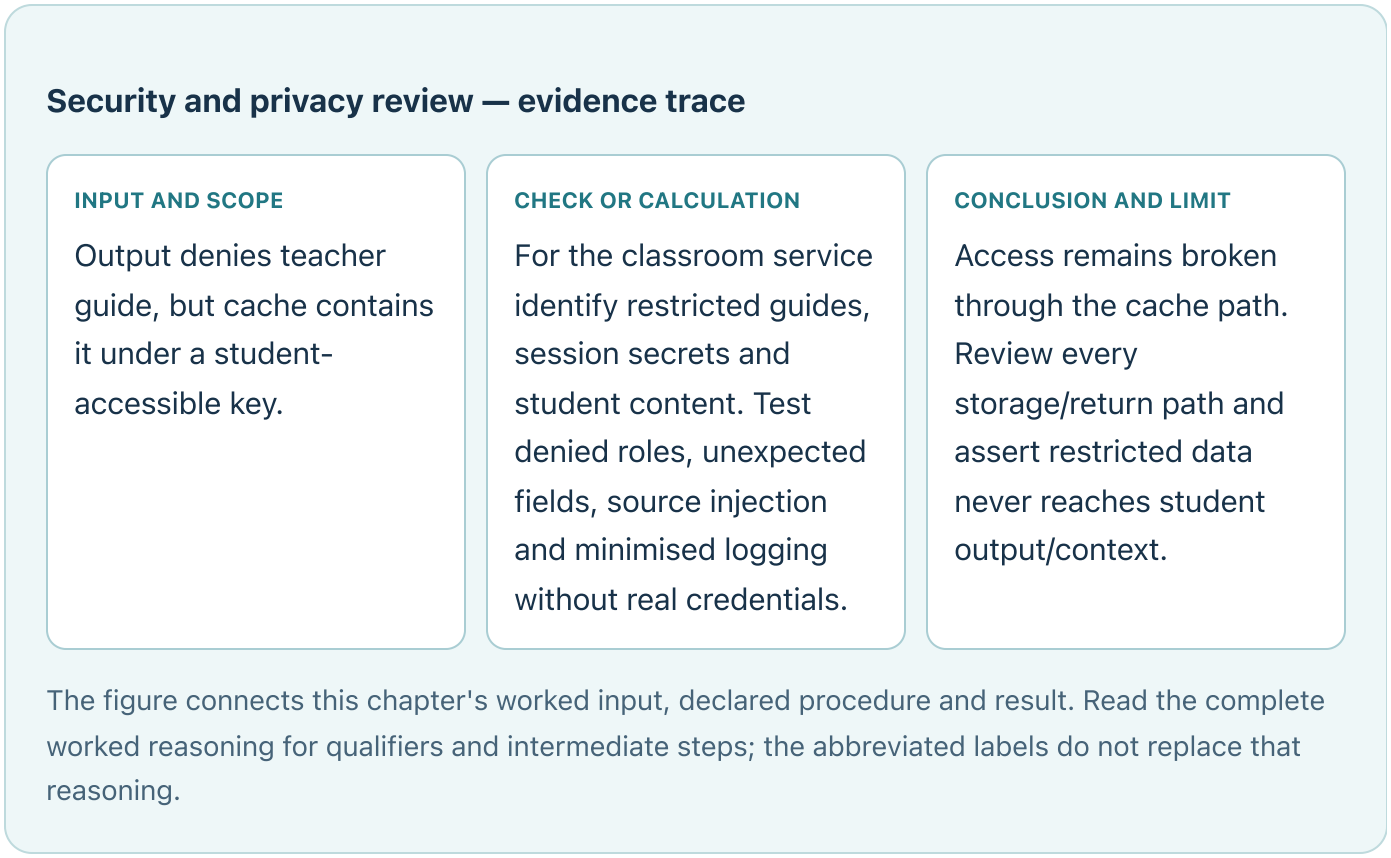
Asset/path	Complete fictional classroom policy
Main restricted guide	G1 contains an obvious fictional marker; teacher-1 may read, learner-1 may not
Trusted identity fixture	Actor/role registry supplied by harness; request-body role claims do not grant access
Main read routes	Direct and cache must apply the same trusted access gate before content lookup/context
Broken route illustration	Teacher guide is stored under a query-only student-accessible cache key
Repaired cache	Scoped teacher key plus current trusted gate; key alone is not authentication
Request schema	Exactly resource_id; unexpected role or other fields rejected
Separate private note N1	Text readable only by learner-1 and assigned teacher-1; teacher-2 is not assigned
Picture analytics	Only revision event kind enters count builder; dashboard gets revision_count, no note text or learner IDs
Privacy limit	Counts may still identify tiny groups; access, linkability and retention need separate review
Scope	Fictional local policy/path probes, not a complete security audit, real login or anonymity guarantee

Main route ledger and checked reasoning

Main declared path/probe	Expected result	Content-read trace
Broken student-accessible query-only cache	Contains restricted marker; direct denial cannot repair it	Unsafe storage/return path remains
Repaired direct route, learner-1 asks G1	Denied	Empty
Repaired cache route, learner-1 asks G1	Denied	Empty
Teacher-1 asks G1 through either repaired route	Allowed fictional guide	One permitted lookup each
Learner request adds role=teacher	Invalid input; no authority upgrade	No added lookup
Unknown trusted actor	Denied	No added lookup

Main direct denial does not protect a guide copied into a student-accessible cache. Both repaired direct/cache routes gate trusted actor before lookup: learner denied with empty read trace, teacher-1 allowed. Unexpected role field rejects; unknown actor denies. Picture note text belongs only to learner-1/assigned teacher-1, not every teacher; teacher-2 denies.

Two revision-kind events produce count 2, while full-note event rejects under counts-only policy. Counts are not automatically anonymous. Code tests declared paths only; injection/retention and other routes remain untested. Selected export bypass needs an authorized teacher-only export or minimized metadata route, plus student-denial tests across every export endpoint/context without real restricted data.



Guided: Access remains broken through the cache path. Review every storage/return path and assert restricted data never reaches student **output**/context.

Qualification: Review findings are bounded by tested paths and threat assumptions, not a proof of complete security.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Output denies teacher guide, but cache contains it under a student-accessible key.

Access remains broken through the cache path. Review every storage/return path and assert restricted data never reaches student **output**/context.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s. The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain only authorized, minimized **evidence** without unnecessary private content so a later revision can be compared honestly.

Misuse, assignment and linkability probes

Review probe	Expected interpretation
Direct request denied but cache returns guide	Boundary still broken; inspect both routes
Unauthorized content read then hidden in output	Already crossed context/read boundary; final redaction alone is insufficient
Request supplies role=teacher	Reject unexpected authority claim; use trusted actor fixture
Teacher not assigned to private note	Deny relationship-scoped access
Analytics event includes note text	Reject under counts-only policy
Counts omit names	Not necessarily anonymous; tiny groups/linkability still matter
Source text contains an instruction	Evidence content does not grant extra authority; never execute it as policy
Only happy-path request tested	Misuse and denied paths remain untested
Local fixtures pass	Scoped policy evidence, not complete security or real identity verification

Separate note-reader and analytics policy

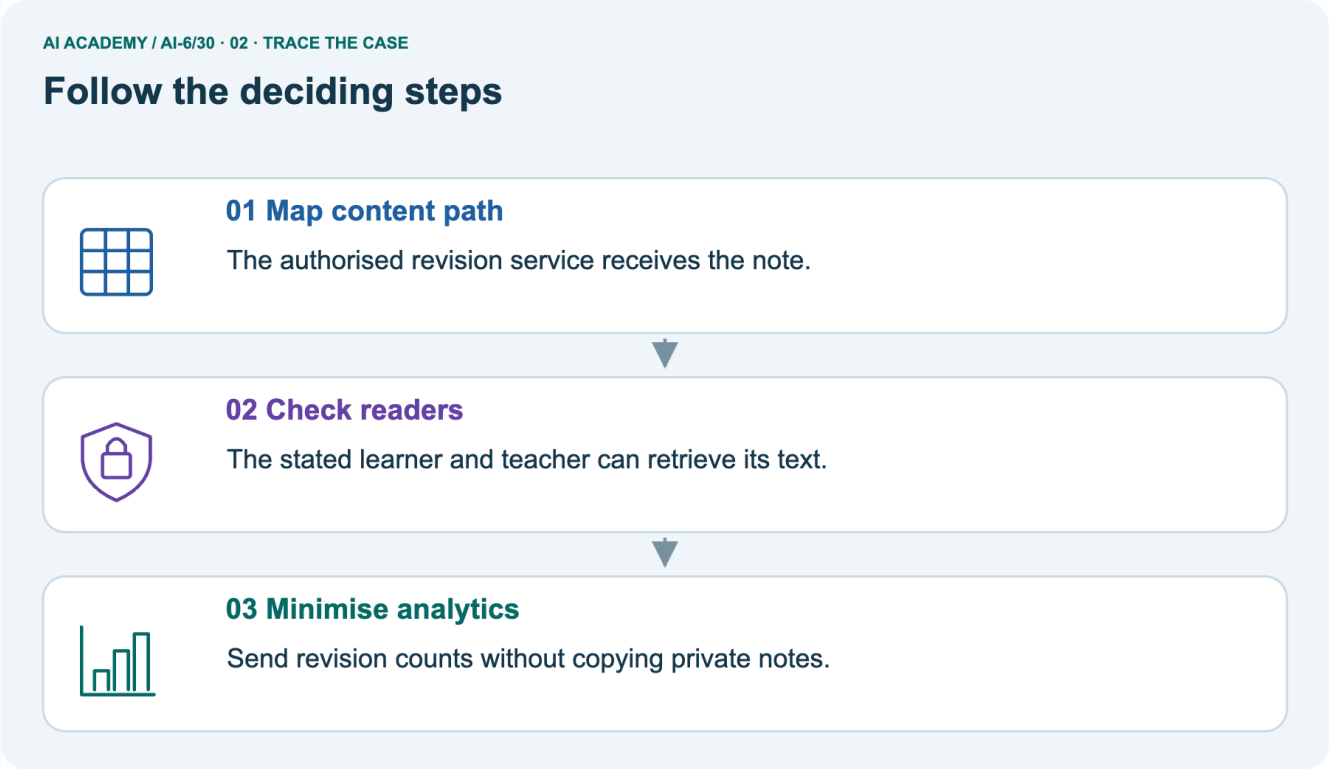
Review a private note’s path. A revision service reads a learner’s private practice note; analytics need counts only.

Only the learner and assigned teacher may read note text. The dashboard is authorised for aggregate counts, not notes.

The picture is a separate private-note path. Its policy permits note text only for the learner and assigned teacher; a broad teacher role alone is insufficient. Analytics is authorized for aggregate revision counts, not text. The added code accepts two fictional revision-kind events and emits revision_count = 2 without note text or learner IDs. A changed event containing the full note is rejected.

This is data minimization under the stated policy, not proof of anonymity: a tiny-group count may still reveal information. The dashboard's readers, group sizes, linkability and retention remain separate review questions.

Picture trace and full-note dashboard key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Map content path	The authorised revision service receives the note.
2. Check readers	The stated learner and teacher can retrieve its text.
3. Minimise analytics	Send revision counts without copying private notes.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **privacy review** follows data and authority through components. A useful dashboard does not automatically need every underlying private detail.

Tiny-group aggregates can still reveal individuals; evaluate linkability, access and retention.

Change: The dashboard starts storing full notes.

Prediction: Flag the unapproved content path and remove it from the design.

Reason: Its count-only purpose does not grant note-reading permission.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Threat model	Assets, actors and failure assumptions considered
Attack surface	The inputs and routes through which behaviour can be influenced
Privacy review	A check of purpose, access and data handling

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

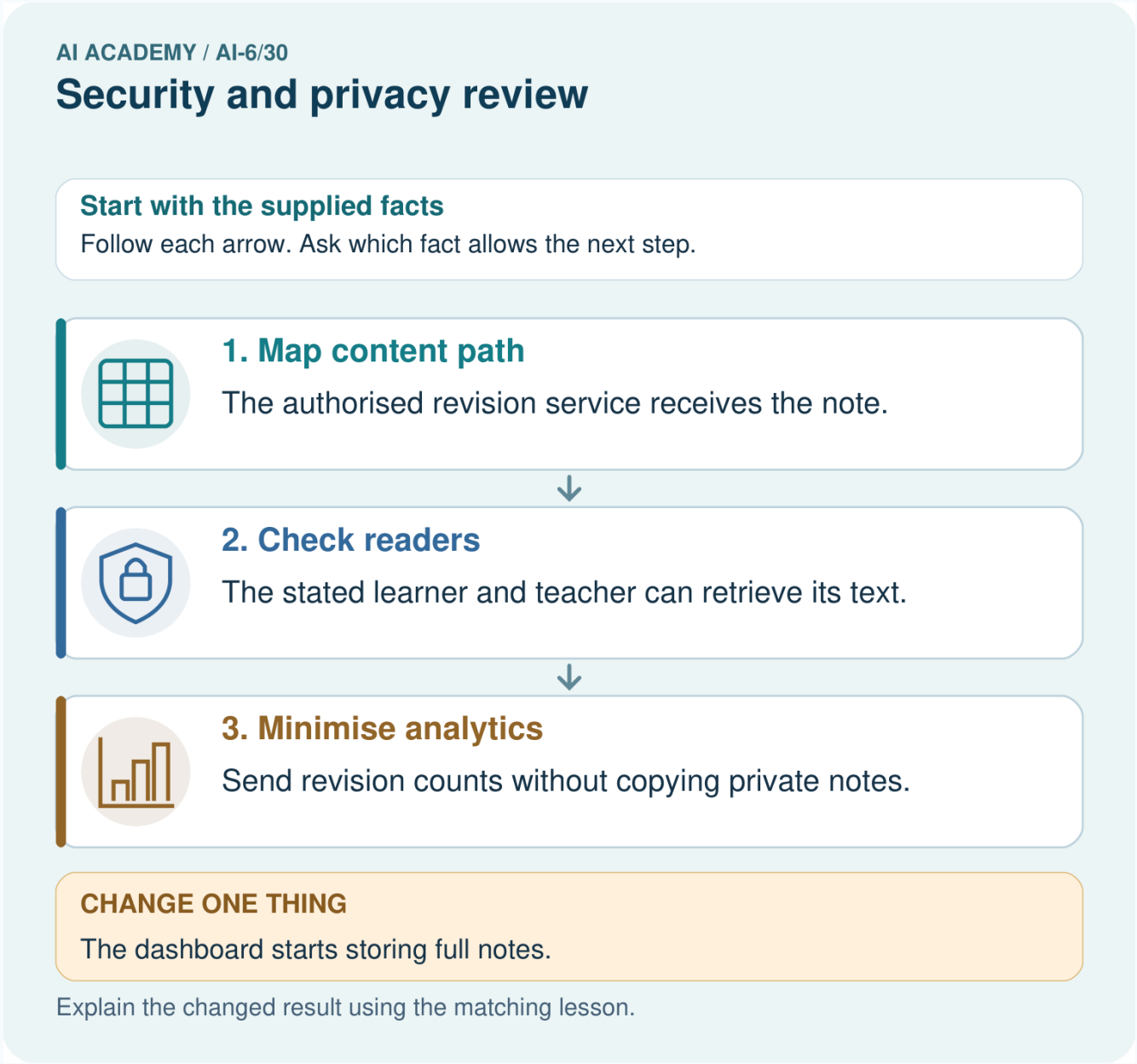
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Security and privacy review

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: **Output** denies teacher guide, but cache contains it under a student-accessible key. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Access remains broken through the cache path. Review every storage/return path and assert restricted data never reaches student **output**/context.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Security and **privacy review**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Review findings are bounded by tested paths and threat assumptions, not a proof of complete security.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Trace private note content and analytics counts to their permitted readers. Why is storing full notes in the dashboard outside this design, and what risk can remain for tiny-group counts?

The authorised revision service receives the note. The stated learner and teacher can retrieve its text. Send revision counts without copying private notes.

Changed case: Flag the unapproved content path and remove it from the design. Its count-only purpose does not grant note-reading permission. Tiny-group counts can remain identifying, so minimise and review them too.

Changed-case reasoning: Flag the unapproved content path and remove it from the design. Its count-only purpose does not grant note-reading permission.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main confidentiality remains broken through the student-accessible cache despite direct denial. Gate both repaired paths before content lookup and inspect empty denied read traces. Picture counts-only analytics rejects note text, and assigned-teacher access differs from any-teacher access; tiny-group counts may still identify people.

Selected diagnostic export bypasses the same boundary: use an authorized teacher-only export or minimized metadata and test student-denied access across every export route. Do not use real restricted content or claim complete review from these fixtures.

Diagnose the current misconceptions

The worked result establishes every future case: Review findings are bounded by tested paths and threat assumptions, not a proof of complete security.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Service blocks direct teacher lookup but diagnostic export includes full teacher answers. Identify defect and a test.

Teacher explanation: Diagnostic export bypasses confidentiality. Use a teacher-only authorised export or minimised metadata; test student requests against every export route.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Security and **privacy review** checks assets, actors, data flows and permissions against explicit threats. A clean happy path does not test misuse.

For the classroom service identify restricted guides, session secrets and student content. Test denied roles, unexpected fields, source injection and minimised logging without real credentials.

W2 Trace the worked case

Output denies teacher guide, but cache contains it under a student-accessible key. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Access remains broken through the cache path. Review every storage/return path and assert restricted data never reaches student **output**/context.

W3 Audit the limit

A report omits this qualification: Review findings are bounded by tested paths and threat assumptions, not a proof of complete security.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Review findings are bounded by tested paths and threat assumptions, not a proof of complete security.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Review findings are bounded by tested paths and threat assumptions, not a proof of complete security.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Access remains broken through the cache path. Review every storage/return path and assert restricted data never reaches student **output**/context.

Qualification: Review findings are bounded by tested paths and threat assumptions, not a proof of complete security.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Service blocks direct teacher lookup but diagnostic export includes full teacher answers. Identify defect and a test.

Teacher explanation: Diagnostic export bypasses confidentiality. Use a teacher-only authorised export or minimised metadata; test student requests against every export route.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Diagnostic export bypasses confidentiality. Use a teacher-only authorised export or minimised metadata; test student requests against every export route. Require intermediate reasoning and the stated scope.

Fresh retry: Secrets hidden from responses but included in error logs. Expected reasoning: Logging leaks the protected asset. Check logs, access and retention as well as response **output**.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled security and **privacy review evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of security and **privacy review**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Red team exercises. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/31, ai-6/33, ai-6/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of security and **privacy review**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: **Output** denies teacher guide, but cache contains it under a student-accessible key. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Access remains broken through the cache path. Review every storage/return path and assert restricted data never reaches student **output**/context.

Check every learner before discussion: Use the worked case for Security and **privacy review**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Review findings are bounded by tested paths and threat assumptions, not a proof of complete security.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The dashboard starts storing full notes. Require a **prediction** and a reason before comparing. Expected result: Flag the unapproved content path and remove it from the design.

Reason: Its count-only purpose does not grant note-reading permission.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

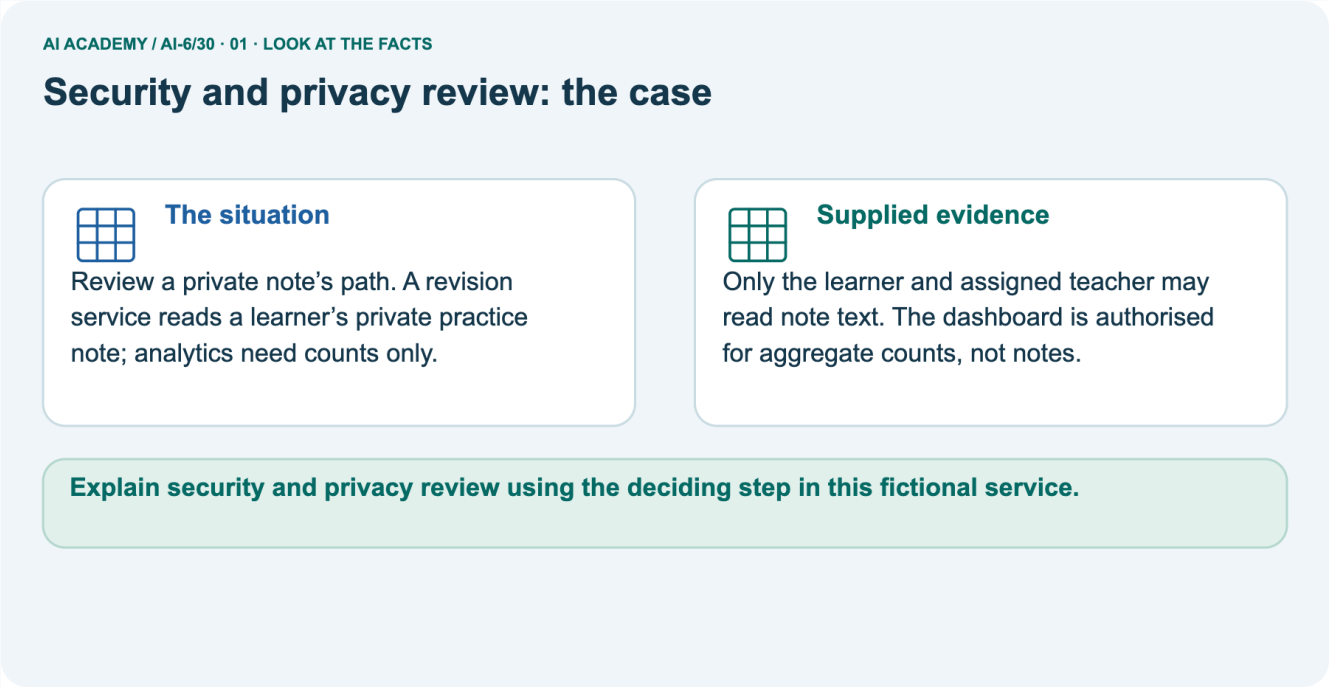
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Security and privacy review: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

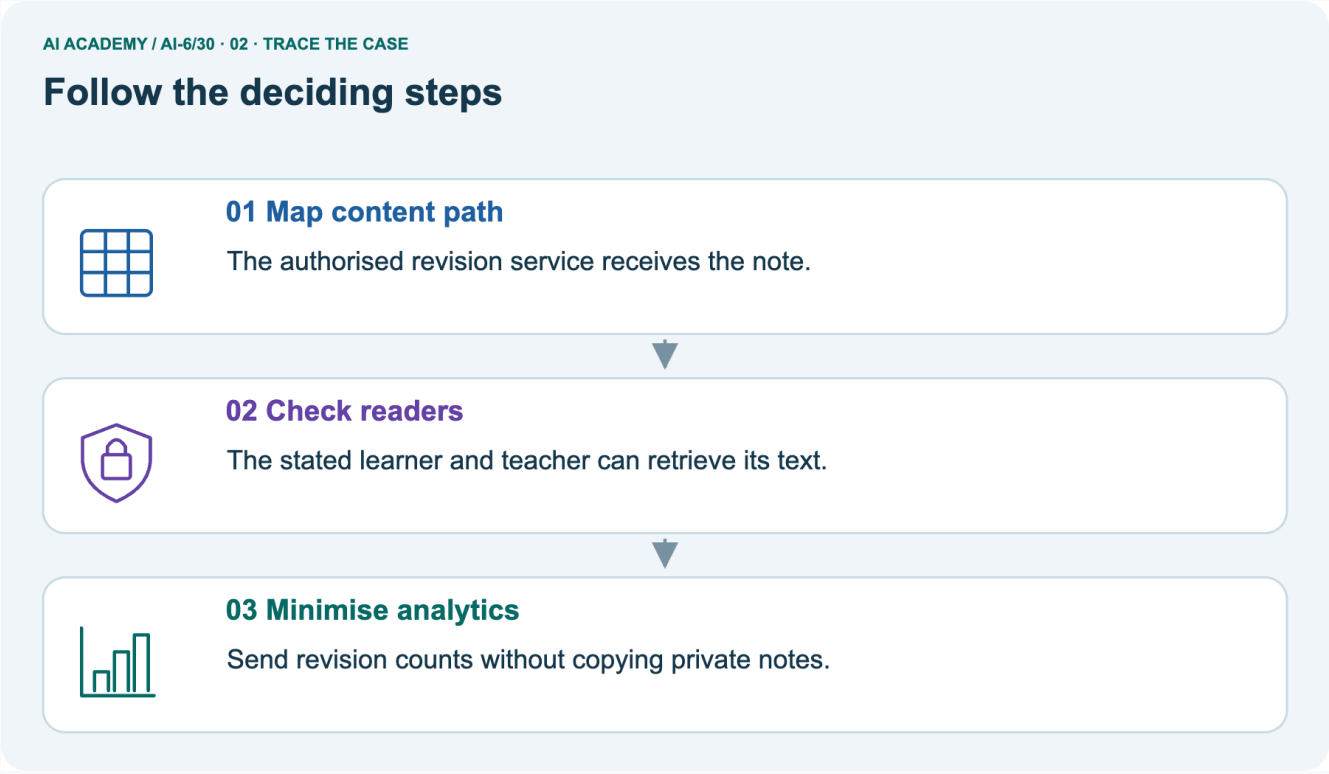
Review a private note's path. A revision service reads a learner's private practice note; analytics need counts only. Supplied facts: Only the learner and assigned teacher may read note text.

The dashboard is authorised for aggregate counts, not notes.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



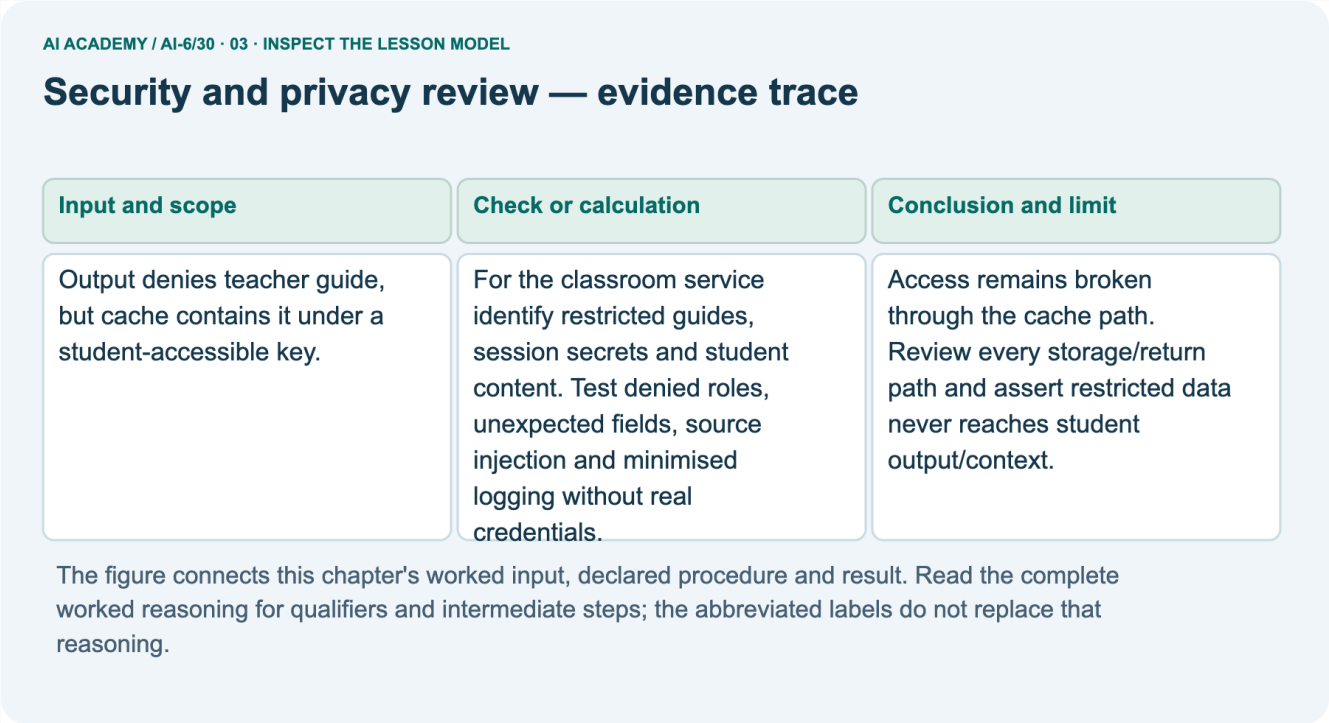
Follow the labelled picture with the matching explanation. Open the vector diagram

Map content path: The authorised revision service receives the note. → Check readers: The stated learner and teacher can retrieve its text. → Minimise analytics: Send revision counts without copying private notes.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Security and privacy review — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Security and **privacy review** — **evidence** trace. **Output** denies teacher guide, but cache contains it under a student-accessible key. | For the classroom service identify restricted guides, session secrets and student content.

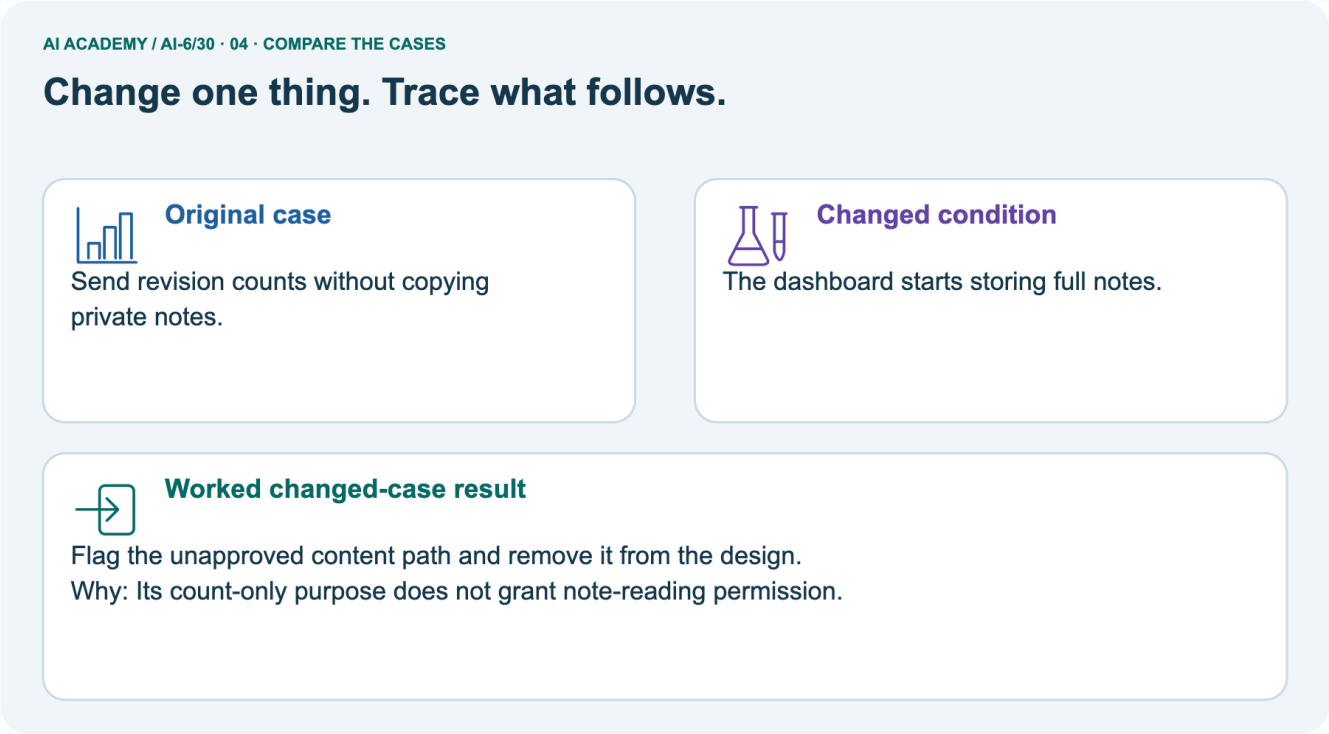
Test denied roles, unexpected fields, source injection and minimised logging without real credentials. | Access remains broken through the cache path.

Review every storage/return path and assert restricted data never reaches student **output**/context.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Send revision counts without copying private notes. Changed condition: The dashboard starts storing full notes. Expected result: Flag the unapproved content path and remove it from the design.

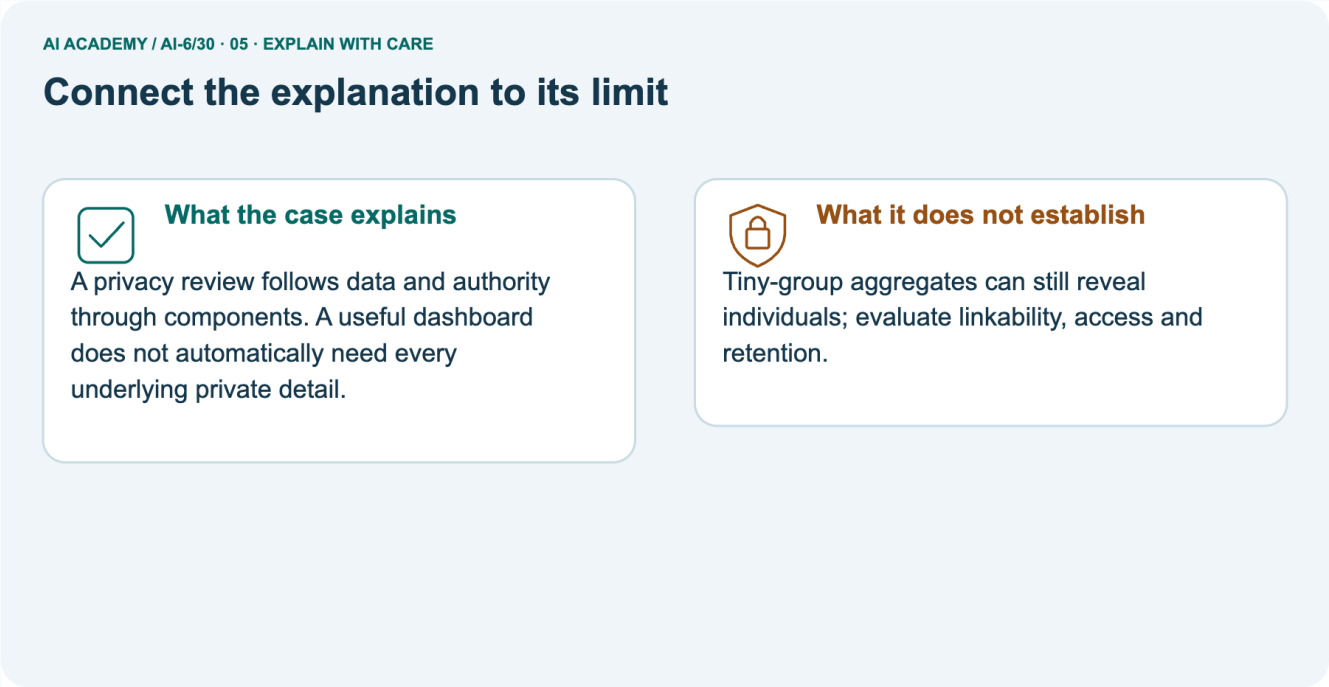
Reason: Its count-only purpose does not grant note-reading permission.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Its count-only purpose does not grant note-reading permission.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **privacy review** follows data and authority through components. A useful dashboard does not automatically need every underlying private detail.

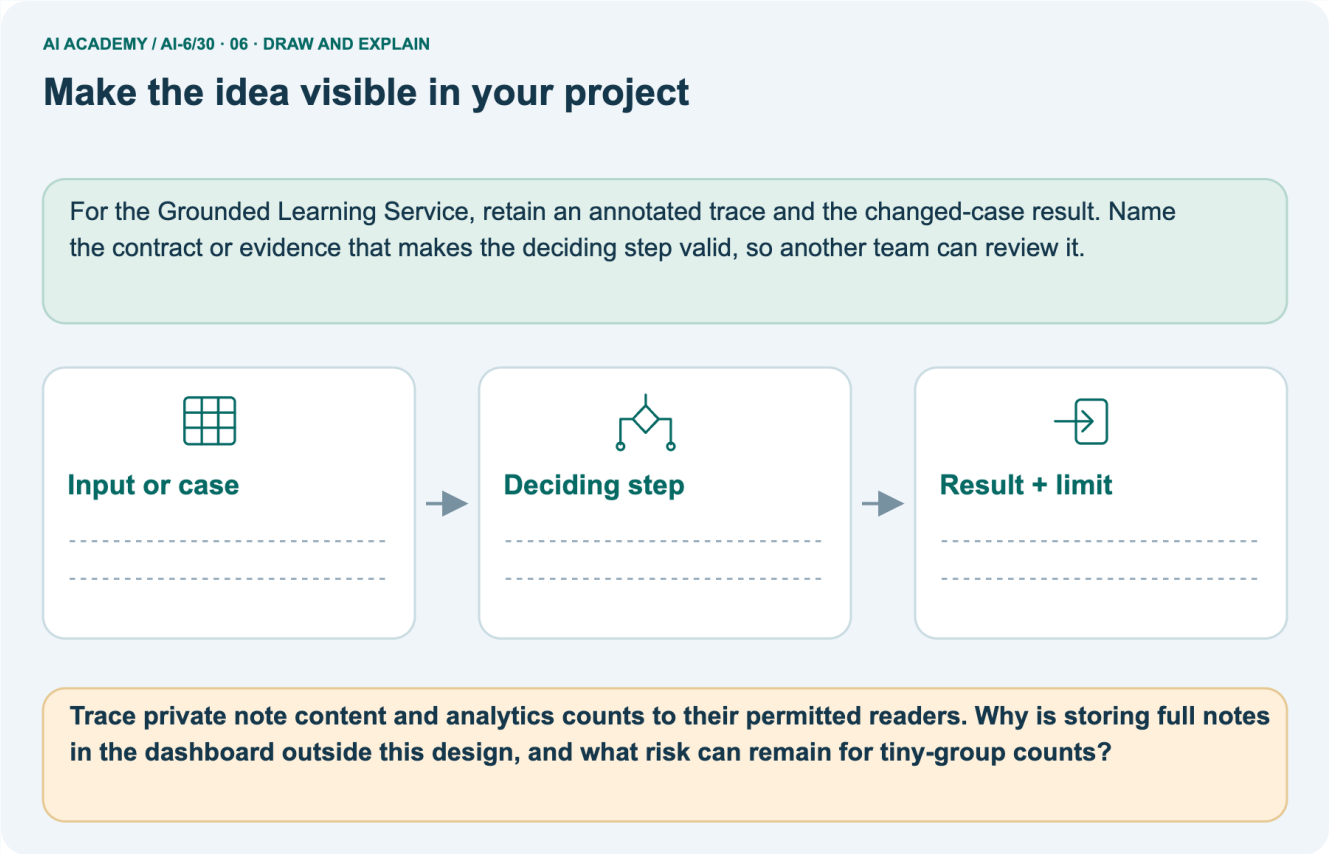
Limit: Tiny-group aggregates can still reveal individuals; evaluate linkability, access and retention.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Tiny-group aggregates can still reveal individuals; evaluate linkability, access and retention.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Trace private note content and analytics counts to their permitted readers.

Why is storing full notes in the dashboard outside this design, and what risk can remain for tiny-group counts?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The authorised revision service receives the note. The stated learner and teacher can retrieve its text. Send revision counts without copying private notes.

Changed case: Flag the unapproved content path and remove it from the design. Its count-only purpose does not grant note-reading permission. Tiny-group counts can remain identifying, so minimise and review them too.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

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▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.